

Linear regression (running + interpreting)

CRISP R Mini-Course

2026

Day 6

Review from last time

What does the code below do? Explain to your neighbor!

```
library(dplyr)
```

```
df <- read.csv("data.csv")
```

```
df_new <- df %>%
```

```
  mutate(sex = factor(sex, labels = c("Female", "Male")),
```

```
         disease = factor(disease, labels = c("no", "yes"))) %>%
```

```
  filter(bmi < 30)
```

```
tx_disease_table <- table(df_new %>% select(sex, disease))
```

```
prop.test(tx_disease_table)
```

↳ test for equal proportions

chi_sq() & handles 2x3

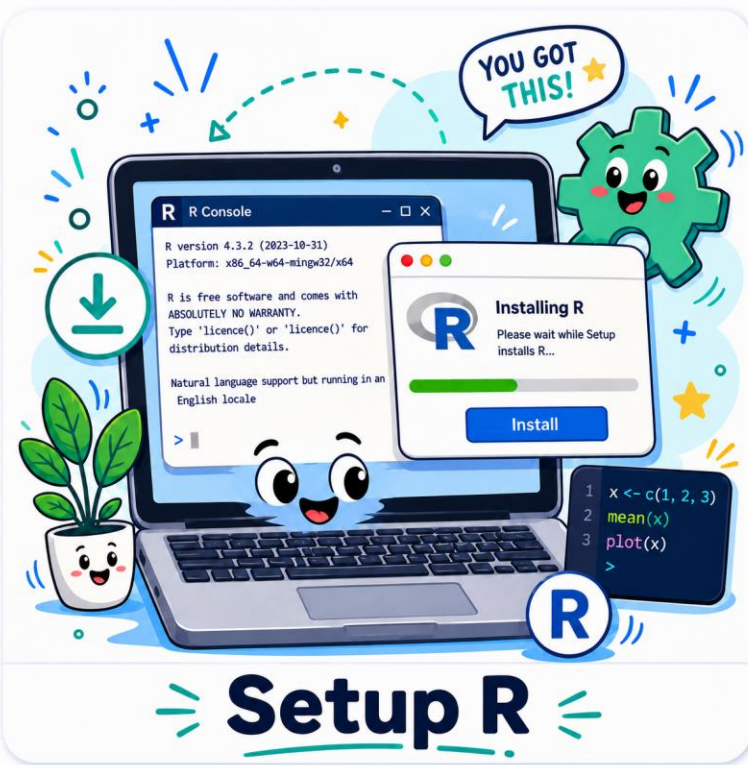
↙ Convert
to factor
variables

↙ Creates
2x2 table

CRISP R course roadmap (1)

Step 0:

Setup R



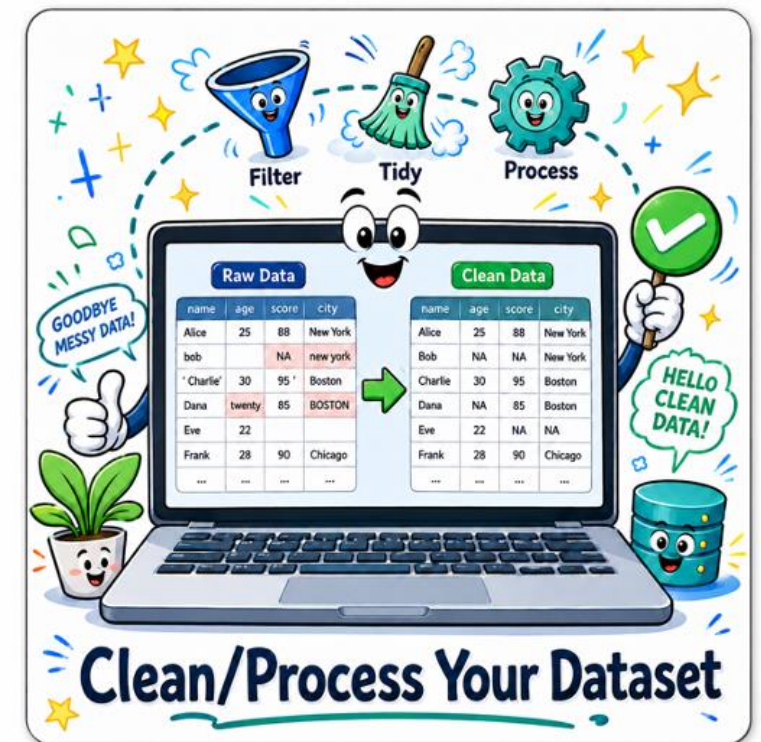
Step 1:

Read in dataset



Step 2:

Clean/process your dataset



CRISP R course roadmap (2)

Step 3:

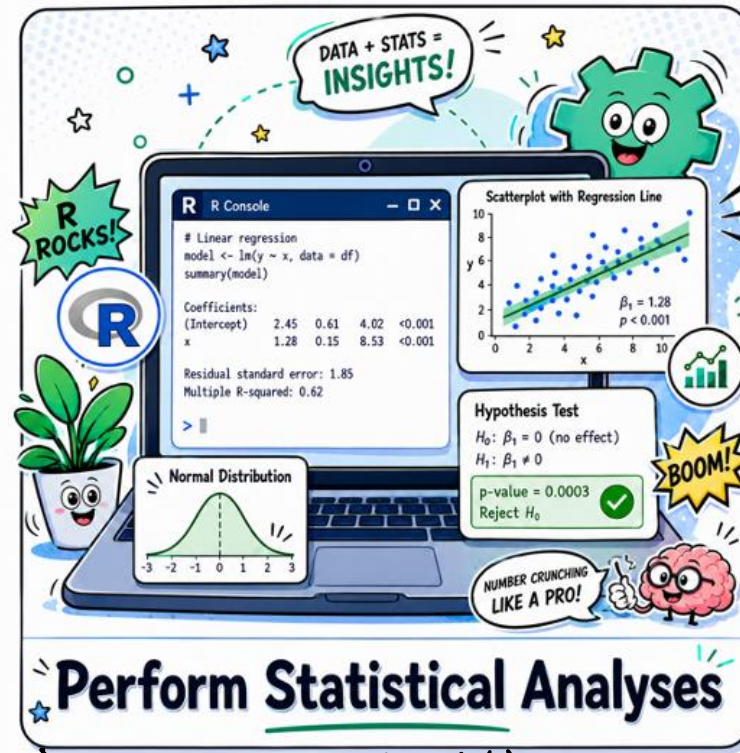
Obtain descriptive statistics



- 1) `group_by()` [`summarise()`] `dplyr`
- 2) `table1` `table1`

Step 4:

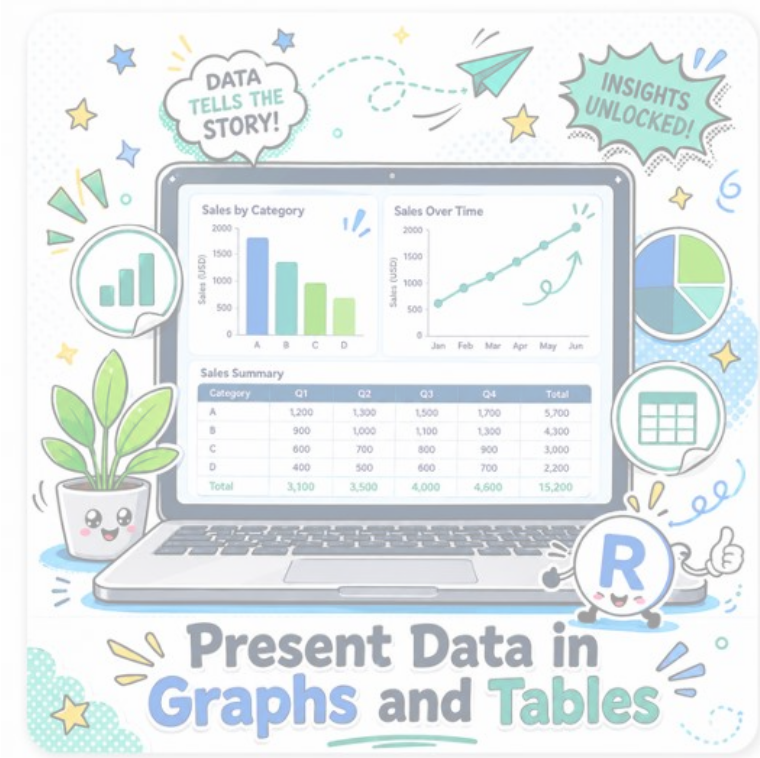
Perform statistical analyses



- 1) compare means - `t.test()`
- 2) compare proportions - `prop.test()` [`chi.sq()`]

Step 5:

Present data in graphs and tables



Today's agenda

- Linear regression – conceptual tutorial
- Running and interpreting linear regression in R

Guided tutorial

Today, we will learn the basics of linear regression.

1. Go to bit.ly/crisp2026.
2. Download (1) new CRISP survey data file, and (2) Rmd file for today into your CRISP R notes folder.

Linear regression w/ continuous variable (1)

- We are trying to determine if ~~#~~ years of education is associated with age. We obtain data from an observational study with both variables.
- We answer this question using linear regression:
 - **Outcome:** Years of education
 - **Exposure:** Age

Linear regression w/ continuous variable (2)

age	educyrs
12	7
15	8

- Linear regression:

- **Outcome:** Years of education

- **Exposure:** Age

Coefficients:

$\ln(\text{educyrs} \sim \text{age}, \dots)$

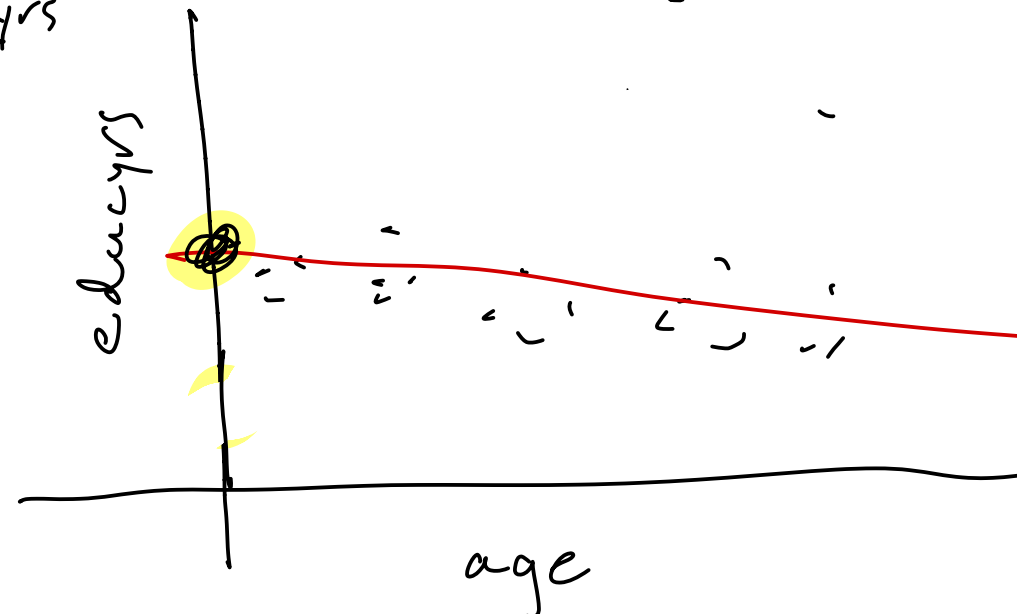
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14.30130	0.77132	18.541	< 2e-16 ***
age	-0.06426	0.02275	-2.825	0.00573 **

Interpretation:

↑ used to
estimate CIs

β_0 : estimated years of education
when age = 0

β_1 : estimated difference in years
of education comparing two groups
one year age apart



$$\text{educyrs} = \beta_0 + \beta_1 \cdot \text{age}$$

p-value: $H_0: \beta_1 = 0$

Reject the null
find a significant
association between age
and educyrs

Linear regression w/ binary variable (1)

- We are trying to determine if a **number of years of education** is associated with **sex**. We obtain data from an observational study with both variables.
- *Q: What is one way we can analyze our data to answer this question?*

Linear regression w/ binary variable (2)

- We are trying to determine if a **number of years of education** is associated with **sex**. We obtain data from an observational study with both variables.
- We can also analyze this using linear regression:
 - **Outcome:** **Number of years of education**
 - **Exposure:** **Sex**

Linear regression w/ binary variable (3)

- Linear regression:

- Outcome:** Years of education

- Exposure:** Sex

$\ln(\text{educyrs} \sim \text{sex})$

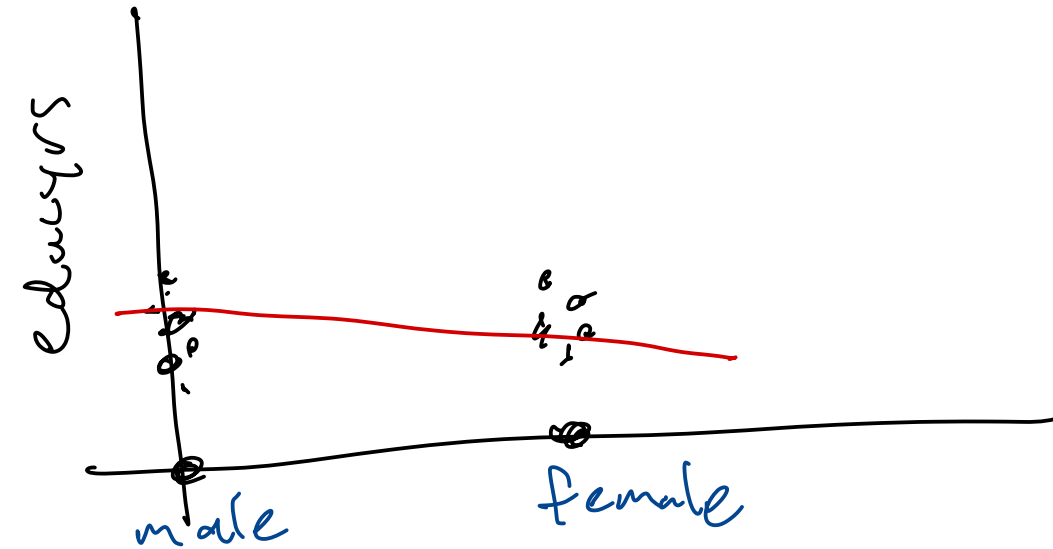
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.21127	0.25203	48.451	<2e-16 ***
sexfemale	-0.03885	0.46801	-0.083	0.934

Interpretation

β_0 : mean years of education in males - 12.21 years

β_1 : difference in the mean years of education comparing males and females



$$\text{educyrs} = \beta_0 + \beta_1 \cdot \text{female}$$

p-value: $H_0: \beta_1 = 0$
no significant difference

Linear regression w/ adjustment variable (1)

- We are trying to determine if **number of years of education** is associated with **age** adjusting for **sex**. We obtain data from an observational study with all variables.
- We answer this question using linear regression:
 - **Outcome:** **Number of years of education**
 - **Exposure:** **Age**
 - **Adjustment covariate:** **Sex**

Linear regression w/ adjustment variable (2)

- Linear regression:
 - **Outcome:** Number of years of education
 - **Exposure:** Age
 - **Adjustment covariate:** Sex

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	14.36048	0.79808	17.994	< 2e-16	***
age	-0.06482	0.02292	-2.828	0.00569	**
sexfemale	-0.14054	0.45358	-0.310	0.75733	